

ABC MANUFACTURING CORPORATION

Environmental Health & Safety Department

STANDARD OPERATING PROCEDURE

Compressed Gas Cylinder Safety

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A. Document Control

A.1 Revision History

Rev	Date	Description of Change	Author	Approver
00	08 Apr 2019	Initial issue of the compressed gas cylinder safety procedure.	J. Alvarez	M. Okafor
01	22 Jul 2023	Added fuel-gas/oxidizer segregation distances and regulator leak-check method.	R. Nguyen	M. Okafor
02	15 Jan 2026	Biennial review. Added risk matrix, gas compatibility chart, cylinder inspection checklist, and asphyxiation controls.	J. Alvarez	M. Okafor

A.2 Distribution List

Copy No.	Holder / Role	Location	Format
Master	Document Owner — Manager, EHS	DMS (electronic master)	Electronic
01	Production Floor Supervisors	Line 1–4 supervisor stations	Controlled print
02	Gas Store / Receiving Lead	Cylinder storage area	Controlled print
03	Welding / Fabrication Lead	Fabrication shop	Controlled print
04	Maintenance Planner	Maintenance office	Electronic
05	Training Coordinator	Learning Management System	Electronic

A.3 Controlled Copy Statement

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1. Purpose

This Standard Operating Procedure establishes the requirements for the safe receipt, handling, transport, storage, use, and return of compressed and liquefied gas cylinders at ABC Manufacturing Corporation. Its purpose is to control the hazards of high-pressure cylinders — impact and “rocketing” from cylinder or valve failure, fire and explosion, asphyxiation, and toxic or corrosive gas exposure — through proper securing, segregation, and use.

- Ensure cylinders are secured upright, capped when not in use, and never allowed to fall or be struck.
- Ensure oxidizing and fuel gases are segregated, and incompatible gases are separated in storage and use.
- Ensure regulators and connections are correct, leak-tight, and free of oil/grease on oxygen service.
- Support the site ISO 45001 OH&S and ISO 14001 environmental management systems.

2. Scope

This procedure applies to all ABC Manufacturing employees and contractors who receive, move, store, connect, or use compressed or liquefied gas cylinders — flammable, oxidizing, inert, toxic, or corrosive — at any ABC Manufacturing facility.

2.1 Inclusions

- Individual industrial cylinders and small manifolds/carts; receiving, transport, storage, securing, regulator connection, use, and empty return.

2.2 Exclusions

- Bulk gas systems, piped gas distribution, and bulk cryogenic tanks (Process Safety and facilities programs).
- Oxygen-fuel gas welding and cutting operations, which additionally follow SOP-EHS-007.
- Cylinder use as a process reactant under a dedicated process SOP, which governs where more stringent.

3. References

This procedure is based on publicly available regulations and consensus standards. Where a conflict exists, the most stringent applicable requirement governs.

Reference	Title / Subject
29 CFR 1910.101	Compressed gases (general requirements)
29 CFR 1910.102 / .103 / .104	Acetylene; hydrogen; oxygen
29 CFR 1910.253	Oxygen-fuel gas welding and cutting — cylinder storage/segregation
NFPA 55	Compressed Gases and Cryogenic Fluids Code
NFPA 51 / 51B	Oxygen-fuel gas systems; fire prevention in cutting/welding
NFPA 704	Identification of the Hazards of Materials for Emergency Response
CGA P-1	Standard for Safe Handling of Compressed Gases in Containers
29 CFR 1910.1200	Hazard Communication — SDS for gases
ISO 45001:2018 / ISO 14001:2015	OH&S and environmental management systems
Internal	SOP-EHS-001 Handling & Storage; SOP-EHS-007 Welding/Cutting/Brazing; SOP-EHS-013 Incident Reporting

4. Definitions

Term	Definition
Compressed Gas	A gas or mixture in a container under pressure, or a liquefied/dissolved gas, meeting the regulatory pressure

Term	Definition
	thresholds.
Valve Protection Cap	The threaded cap that protects the cylinder valve from damage; must be in place whenever a regulator is not attached.
Regulator	A device that reduces cylinder pressure to a usable working pressure; must match the gas and CGA connection.
CGA Connection	The standardized valve outlet connection assigned to a gas to prevent cross-connection of incompatible gases.
Fuel Gas	A flammable gas such as acetylene, propane, or hydrogen used for heating, cutting, or welding.
Oxidizer	A gas such as oxygen or nitrous oxide that supports and intensifies combustion.
Asphyxiant	An inert gas (e.g., nitrogen, argon, helium) that can displace oxygen and cause suffocation without warning.
Empty / "MT"	A cylinder taken out of service; retains residual pressure, must be closed, capped, marked, and stored separately.

5. Responsibilities

Role	Responsibility
Area Supervisor	Ensures only trained personnel handle cylinders, verifies securing and segregation, authorizes cylinder use.
Cylinder Handler / Operator	Inspects, transports, secures, and connects cylinders per this SOP; performs leak checks; reports damaged cylinders.
Gas Store / Receiving Lead	Verifies cylinder condition and labeling at receipt, maintains segregation and inventory, manages empty returns.
Maintenance	Maintains carts, restraints, storage racks, ventilation, and gas-detection equipment.
EHS	Approves gas additions, audits storage/segregation, supports incident investigation.

6. Required Training

Training Module	Audience	Frequency
Compressed Gas Cylinder Safety (this SOP)	All cylinder handlers	Initial + on revision
Cylinder transport & securing	All cylinder handlers	Initial + annual
Regulator connection & leak check	Operators connecting cylinders	Initial + 2-yr refresh
Gas hazards & asphyxiation awareness	All personnel in gas-use areas	Initial + annual
Fuel-gas/oxidizer segregation	Gas store, welding leads	Initial + 2-yr refresh

7. Required Qualifications

- Completion of the training in Section 6, verified in the LMS, and authorization by the Area Supervisor.
- Demonstrated competency in cylinder securing, regulator connection, and leak checking during on-the-job evaluation.
- Awareness of oxygen-deficient/asphyxiation hazards and the prohibition on entering to rescue without proper controls.

8. Required PPE

PPE is selected for the gas and task from the SDS. The following is the minimum baseline for cylinder handling.

Task	Eye/Face	Hand	Foot	Other
Transport / securing	Safety glasses	Leather/gripping gloves	Safety shoes (toe	—

Task	Eye/Face	Hand	Foot	Other
			protection)	
Regulator connect / leak check	Safety glasses	Nitrile/leather gloves	Safety shoes	—
Corrosive/toxic gas handling	Goggles + face shield	Chemical-resistant gloves	Safety shoes	Gas detection / respiratory per SDS
Inert gas in confined/low areas	Safety glasses	Gloves	Safety shoes	O2 monitor; confined-space controls

WARNING

Never use oil, grease, or hydrocarbon-contaminated gloves/tools on oxygen cylinders, valves, or regulators. Oxygen can react violently with hydrocarbons and cause fire or explosion.

9. Required Equipment

- Cylinder hand truck/cart designed to secure the cylinder during transport; chains, straps, or racks to secure cylinders upright in storage and use.
- Correct regulator and CGA connection for each gas; cylinder valve protection caps; appropriate valve/cap wrench (no cheater bars).
- Approved leak-detection solution (compatible; oil-free for oxygen); never a flame.
- Ventilation and fixed/portable oxygen (O2) monitors where inert or asphyxiant gases are used; toxic-gas detection where applicable.
- NFPA 704 placarding, segregation barriers/spacing, and fire extinguisher appropriate to the area.

10. Hazard Identification

- **Impact / “rocketing”** — a dropped cylinder or sheared valve releasing stored pressure can turn the cylinder into a projectile.
- **Fire / explosion** — fuel gas leaking near an ignition source, or oxygen enrichment intensifying combustion.
- **Asphyxiation** — inert gas (nitrogen, argon, helium) displacing oxygen in low or confined areas, without warning.
- **Toxic / corrosive exposure** — release of a toxic or corrosive gas causing acute health effects.
- **Oxygen enrichment** — elevated oxygen making clothing and materials highly flammable.
- **Struck-by / crush** — heavy cylinders during manual handling.

11. Risk Assessment

Risk level is derived from the severity × likelihood matrix. Any residual risk assessed as High or Critical requires additional controls and Area Supervisor authorization before work proceeds.

11.1 Risk Rating Matrix

Severity ↓ / Likelihood →	Rare	Unlikely	Possible	Likely	Almost Certain
Catastrophic	Medium	High	High	Critical	Critical
Major	Medium	Medium	High	High	Critical
Moderate	Low	Medium	Medium	High	High
Minor	Low	Low	Medium	Medium	High
Negligible	Low	Low	Low	Low	Medium

11.2 Task Risk Register

Hazard	Severity	Likelihood	Risk Level	Control Measures
Cylinder falls / valve shears (rocketing)	Catastrophic	Unlikely	High	Secure upright, cap on when not in use, cart for transport, never drop/drag, handle by body not cap.
Fuel gas + oxidizer proximity fire	Catastrophic	Unlikely	High	Segregate O2 from fuel gas by 20 ft or a 5-ft ½-hr fire-rated barrier; no ignition sources.
Asphyxiation from inert gas	Major	Possible	High	Ventilation, O2 monitoring, no confined use without controls, never enter to rescue unprotected.
Oil/grease on oxygen equipment	Catastrophic	Rare	Medium	Keep oxygen service oil-free; dedicated clean regulators; no hydrocarbon contact.
Toxic/corrosive gas leak	Major	Unlikely	Medium	Leak check on connection, gas detection, correct CGA fitting, SDS PPE.
Manual handling injury	Moderate	Likely	High	Cart/mechanical aids, correct technique, foot protection.

12. Detailed Step-by-Step Procedure

12.1 Pre-Job Preparation

1. Obtain the SDS for the gas; confirm hazard class (flammable/oxidizer/inert/toxic/corrosive), PPE, and segregation/handling requirements.
2. Confirm the correct regulator and CGA connection for the gas, and that gas detection/ventilation is available where required.

12.2 Receiving & Cylinder Inspection

1. Verify cylinder labeling identifies the gas; do not rely on color alone. Confirm the DOT markings and current hydrostatic test date.
2. Inspect for dents, gouges, arc burns, bulges, corrosion, and valve/thread damage. Reject and tag out any damaged cylinder; return to supplier.
3. Confirm the valve protection cap is present and hand-tight.

12.3 Transport

1. Close the valve and install the valve protection cap before moving any cylinder. Never move a cylinder with a regulator attached.
2. Move cylinders on a cylinder cart secured with chain/strap. Never lift by the cap, drag, roll horizontally (except a short controlled tilt-and-roll), or drop cylinders.

WARNING

Never lift a cylinder by its valve protection cap and never use magnets or slings not designed for cylinders. A dropped cylinder with a sheared valve can become a lethal projectile.

12.4 Storage & Segregation

1. Store cylinders upright, secured by chain/strap or in a rack, with caps on, away from heat, ignition sources, and traffic damage.
2. Segregate oxidizers (e.g., oxygen) from fuel gases (e.g., acetylene) by at least 20 ft, or by a noncombustible barrier at least 5 ft high with a minimum ½-hour fire rating.
3. Store acetylene and other dissolved/liqeuified fuel gases valve-end up. Separate full and empty cylinders and mark empties "MT."
4. Placard storage per NFPA 704 and keep aisles, exits, and fire equipment clear.

12.5 Regulator Connection & Leak Check

1. Confirm the regulator matches the gas/CGA connection; inspect for damage and, for oxygen, confirm oil-free condition. Never force a mismatched fitting or use adapters.
2. “Crack” the valve briefly away from people to clear debris (except toxic/flammable gases where prohibited), then attach the regulator and tighten with the correct wrench.
3. Open the cylinder valve slowly while standing to the side of the regulator, not in front of or behind the gauge faces. For acetylene, open no more than about three-quarters to one and a half turns and leave the wrench on the valve.
4. Leak-check all connections with an approved leak-detection solution — never a flame. Correct any leak before use.

12.6 Use & Verification

1. Use the gas at the intended working pressure; do not exceed the manufacturer's maximum withdrawal rate (important for acetylene).
2. Monitor for leaks and, where inert/asphyxiant gases are used, confirm the oxygen monitor reads normal. Keep cylinders secured during use.

12.7 Cylinder Change-Out & Empty Return / Housekeeping

1. When empty or removed from service, close the valve, relieve regulator pressure, remove the regulator, install the cap, and mark “MT.”
2. Store empties upright, secured, and segregated from full cylinders; stage for supplier return. Never fully empty a cylinder — leave positive residual pressure.

12.8 Documentation

1. Record receiving inspection, storage inspection, and leak checks (Appendices B/C); update the cylinder inventory; file in the DMS.

12.9 Escalation Procedures

1. Damaged cylinder or valve leak that cannot be stopped by tightening → tag out, isolate, ventilate, and notify supervisor/EHS; for flammable/toxic gas, treat as emergency.
2. Low oxygen alarm or suspected asphyxiant release → evacuate, do not enter to rescue without proper respiratory protection and controls.

13. Emergency Response

WARNING

Never enter an oxygen-deficient or toxic atmosphere to attempt rescue without SCBA/SAR and trained standby. Untrained rescuers becoming victims is a leading cause of multiple-fatality gas incidents.

13.1 Immediate Actions

- Evacuate the area, eliminate ignition sources for flammable leaks, ventilate if safe, and isolate the cylinder from a safe distance.

13.2 Medical Response

- Move exposed persons to fresh air; for frostbite from liquefied gas or exposure, seek medical aid; provide the SDS to responders.

13.3 Fire Response

- For a fuel-gas fire, if the valve can be closed safely, do so; otherwise evacuate and let the ERT/fire brigade cool surrounding cylinders and manage the fire.

13.4 Leak / Release Response

- Small leak stoppable by closing the valve or tightening the connection — do so if safe; otherwise treat as an emergency and summon the ERT.

13.5 Equipment / Area Shutdown

- Isolate the gas supply and de-energize ignition-capable equipment in the area from a safe location where possible.

13.6 Notification Chain

Step	Notify	Contact / Method	When
1	Area Supervisor	Radio / floor phone	Immediately on leak, fire, or low-O ₂ alarm
2	EHS On-Call	Ext. 2400 / site radio	Within 5 minutes
3	Emergency Response Team	Emergency line 2222	Fire, injury, uncontrolled leak, or asphyxiation risk
4	External Emergency Services	911	Life threat, fire, or major release
5	EHS Manager & Plant Manager	Per call tree	Within 30 minutes; recordable/reportable events

14. Required Documentation

Record	Purpose	Owner	Retention
Cylinder Receiving/Inspection Checklist (App. B)	Verify condition, markings, cap at receipt	Gas Store Lead	3 years
Storage & Segregation Inspection (App. C)	Monthly verification of storage compliance	Area Supervisor	3 years
Regulator Leak-Check Record	Evidence of leak-tight connections	Operator	1 year
Cylinder Inventory Log	Stock and full/empty status	Gas Store Lead	3 years
Training Records (LMS)	Evidence of qualified handlers	Training Coordinator	Duration + 3 years
Incident Report	Leaks, fires, injuries, asphyxiation events	EHS	5 years

15. References (Public Standards)

- OSHA 29 CFR 1910.101 — Compressed gases (general).
- OSHA 29 CFR 1910.102 / .103 / .104 — Acetylene; hydrogen; oxygen.
- OSHA 29 CFR 1910.253 — Oxygen-fuel gas welding and cutting (cylinder storage).
- NFPA 55 — Compressed Gases and Cryogenic Fluids Code; NFPA 51/51B.
- NFPA 704 — Identification of the Hazards of Materials for Emergency Response.
- CGA P-1 — Safe Handling of Compressed Gases in Containers.
- ISO 45001:2018; ISO 14001:2015 — Management-system standards.

16. Appendices

Appendix A — Gas Compatibility / Segregation Chart

Store separately →	Flammable (fuel)	Oxidizer (O ₂)	Inert	Toxic	Corrosive
Flammable (fuel)	OK	NO	OK	CAUTION	CAUTION

Store separately →	Flammable (fuel)	Oxidizer (O2)	Inert	Toxic	Corrosive
Oxidizer (O2)	NO	OK	OK	CAUTION	CAUTION
Inert	OK	OK	OK	OK	OK
Toxic	CAUTION	CAUTION	OK	OK	CAUTION
Corrosive	CAUTION	CAUTION	OK	CAUTION	OK

Segregate oxygen/oxidizers from fuel gases by 20 ft or a 5-ft, ½-hour fire-rated noncombustible barrier. OK = may be co-located with securing; CAUTION = separate per SDS/CGA; NO = must be separated. Always defer to the SDS and CGA guidance.

Appendix B — Cylinder Receiving & Condition Inspection Checklist

#	Inspection Item	Pass	Fail	N/A
1	Gas identified by label/stencil (not color alone)	☐	☐	☐
2	DOT markings present; hydrostatic test date current	☐	☐	☐
3	No dents, gouges, bulges, arc burns, or corrosion	☐	☐	☐
4	Valve and threads undamaged; no leaks	☐	☐	☐
5	Valve protection cap present and hand-tight	☐	☐	☐
6	Correct CGA connection for intended regulator	☐	☐	☐
7	Compatible storage location available (segregation)	☐	☐	☐

Inspected by: _____ Date: _____ Disposition (Accept / Reject / Return): _____

Appendix C — Monthly Storage & Segregation Inspection

#	Item	OK	Action Needed
1	Cylinders upright and secured (chain/strap/rack)	☐	☐
2	Valve caps in place on cylinders without regulators	☐	☐
3	Oxidizers segregated from fuel gases (20 ft / barrier)	☐	☐
4	Full and empty cylinders separated; empties marked "MT"	☐	☐
5	Away from heat/ignition; ventilation adequate	☐	☐
6	O2 monitor(s) functional where inert gases stored/used	☐	☐
7	NFPA 704 placarding correct; aisles/exits clear	☐	☐

Appendix D — Cylinder Handling Decision Flow

START → Is the gas identified by label and the SDS on file? — NO → Do not use; identify before handling.

YES → Is the cylinder undamaged with a current test date and cap present? — NO → Tag out and return to supplier.

YES → For transport: valve closed, cap on, secured on cart? — NO → Correct before moving.

YES → For storage: upright, secured, segregated (O2 from fuel gas)? — NO → Relocate to compliant storage.

YES → For use: correct regulator/CGA, oil-free (O2), leak-checked, ventilation/O2 monitor OK? — YES → Use per Section 12 → END.